Exact Certified Pricing under the Rational Rough Heston Characteristic Exponent: A From-Scratch Re-Derivation and Cross-Document Reconciliation

Reconciliation and re-proof prepared for the arb4j / bonanzai document set

July 4, 2026

Abstract

We re-derive, from first principles and without treating any prior document as authoritative, the complete theory underlying the certified arbitrary-precision evaluation of European option prices under the rough Heston model through its rational (diagonal Padé) characteristic exponent. The pipeline is: the Müntz-basis fractional Riccati coefficient recurrence; the cumulant (lattice) assembly; the Chebyshev–Wheeler orthogonal-polynomial construction; the diagonal Padé representation with Nuttall–Pommerenke, Stahl, and de Montessus convergence; partial-fraction expansion; and the Schwinger–Gauss–erfc–Hermite closed-form price series with a rigorous series-remainder majorant. We prove the central equivalence: the Edgeworth–Hermite series, the single-index erfc–Hermite series, and the pole–residue double sum are three rearrangements of one absolutely summable family and are therefore equal to the Lewis contour price. We prove the exact, run-time-checkable convergence condition for the integrated Edgeworth price series and show that it holds for every maturity for which the model is defined, subject only to a stated inequality. We reconcile the Edgeworth-divergence document with the convergence results by identifying precisely the object (the true non-rational exponent versus the rational Padé exponent) to which each statement applies, and we adjudicate every remaining cross-document contradiction as a proved proposition. Finally we prove the end-to-end theorem: for every requested bit count b, the pipeline returns a ball enclosure of the exact call/put price of radius at most 2^{-b}, with put–call parity and the Black–Scholes limit as exact certificates.

Contents

1 Canonical notation and the resolution of naming conflicts

The two document trees use several symbols in mutually incompatible ways. We fix one canonical convention here and prove, in Section A, that each alternative usage is either equivalent to ours under an explicit substitution or is an error. All later sections use only the canonical column.

Definition 1.1 (Canonical model data)**.** The rough Heston model carries parameters

$$H \in (0, \tfrac{1}{2}), \quad \lambda > 0, \quad \nu > 0, \quad \rho \in (-1, 0], \quad \theta > 0, \quad V_0 > 0,$$

with fractional order $\alpha := H + \frac{1}{2} \in (\frac{1}{2}, 1)$; contract data $K > 0$, $T > 0$, spot $S_0 > 0$, rate $r \geq 0$; and log-moneyness $k := \log(K/S_0) - rT \in \mathbb{R}$. The symbol Γ is the Euler gamma function; $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-t^2}\,\mathrm{d}t$ is the (entire) complementary error function; He_n denotes the probabilist's Hermite polynomials, $\mathrm{He}_n(x) = (-1)^n e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2}$.

Definition 1.2 (Standing polynomials and Fourier variable)**.** The Fourier-conjugate variable to the log-return $X_T = \log(S_T/S_0)$ is $v \in \mathbb{C}$, reserved for that role. The right-hand side of the fractional Riccati equation is carried by

$$P(v) := -\tfrac{1}{2}(v^2 + \mathrm{i}v), \qquad Q(v) := \lambda(\mathrm{i}\rho\nu\, v - 1), \qquad R := \tfrac{1}{2}\lambda^2\nu^2, \tag{1}$$

so that $P \in \mathbb{C}[v]$ has degree 2 with zeros exactly at $v = 0$ and $v = -\mathrm{i}$, $Q \in \mathbb{C}[v]$ has degree 1, and $R \in \mathbb{R}_{>0}$.

Proposition 1.3 (λ and κ denote the same scalar)**.** *The patent's mean-reversion symbol κ in the drift term $\kappa\theta$ and in Q is identical to the λ of (1).*

Proof. Both appear multiplying the deterministic mean-reversion increment $(\theta - V)$ in the variance dynamics $\mathrm{d}V = \lambda(\theta - V)\,\mathrm{d}t + \nu\sqrt{V}\,\mathrm{d}W$ of the (rough) Heston variance process; matching the coefficient of V in the drift of the associated Riccati right-hand side gives the linear coefficient $-\lambda$ in $Q(v) = \lambda(\mathrm{i}\rho\nu v - 1)$. The two symbols occupy the same structural slot and take the same value; the choice is purely typographic. We use λ. □

Provenance. `notation.tex` fixes λ; `patentApplication.tex` uses $\kappa\theta$ in Algorithm 2. Both are correct and denote the same quantity.

Canonical	Meaning	Conflicting usages resolved
λ	mean-reversion speed	Written κ in the patent's drift $\kappa\theta$; same scalar (Prop. 1.3).
ρ	correlation, $\rho \in (-1, 0]$	Distinct from the rational part $\varrho_M(v)$ and from pole separations δ_j.
$\varrho_M(v)$	proper rational part of κ_M	Some documents write this "$\rho(u)$"; never the correlation.
$\alpha = H + \frac{1}{2}$	fractional order	Some documents write "μ" for the order; canonical is α, and μ_T is drift.
μ_T	Gaussian drift datum of κ_M	Not the fractional order.
σ_T^2	Gaussian envelope datum (leading v^2 coefficient of $-\kappa_M$)	Distinct from the variance $\kappa_2^{(M)}(T)$; see Def. 8.1.
$\kappa_2^{(M)}(T)$	second cumulant (variance) of X_T under φ_M	Equals $\sigma_T^2 + \varrho_M''(0)$ in the s-chart.
$a(k, v)$	Müntz coefficients (0-indexed)	1-indexed $a_k = a(k-1, v)$ appears elsewhere; degrees differ by the shift.
$d(k, v)$	lattice (cumulant) coefficients	—
c_n	Edgeworth / Gram–Charlier weight	Distinct from PFE residue A_j and contour abscissa c_L.
A_j, u_j	PFE residue and pole, $j = 1, \ldots, 2M$, $\operatorname{Im} u_j < 0$	—
$\delta_j := -\operatorname{Im} u_j > 0$	pole–contour separation	Written "ρ_j" in the patent; renamed to avoid the correlation clash.
$z = T^\alpha$	Padé variable	The Padé is in z at fixed v, not in v at fixed T (Prop. 6.2).

Table 1: Canonical notation and the conflicts it resolves. Full adjudication in Section A.

2 Fractional-calculus and Müntz preliminaries

Definition 2.1 (Riemann–Liouville fractional integral)**.** For $\beta > 0$ and $f \in L^1_{\mathrm{loc}}(0, \infty)$, the Riemann–Liouville fractional integral of order β is

$$(I^\beta f)(t) := \frac{1}{\Gamma(\beta)} \int_0^t (t-s)^{\beta-1} f(s)\, \mathrm{d}s, \qquad t > 0. \tag{2}$$

Lemma 2.2 (Fractional integral acts diagonally on the Müntz monomials)**.** *For every $\gamma > -1$ and $\beta > 0$,*

$$(I^\beta t^\gamma)(t) = \frac{\Gamma(\gamma+1)}{\Gamma(\gamma+\beta+1)}\, t^{\gamma+\beta}. \tag{3}$$

In particular, on the Müntz scale $\{t^{k\alpha} : k \in \mathbb{N}_0\}$ with $\alpha > 0$ one has $I^\alpha t^{k\alpha} = \dfrac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)}\, t^{(k+1)\alpha}$.

Proof. Substitute $s = t\tau$ in (2):

$$(I^\beta t^\gamma)(t) = \frac{t^{\gamma+\beta}}{\Gamma(\beta)} \int_0^1 (1-\tau)^{\beta-1} \tau^\gamma \, \mathrm{d}\tau = \frac{t^{\gamma+\beta}}{\Gamma(\beta)}\, B(\gamma+1, \beta),$$

where B is the Euler beta integral, convergent because $\gamma + 1 > 0$ and $\beta > 0$. Using $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ gives $(I^\beta t^\gamma)(t) = \frac{\Gamma(\gamma+1)}{\Gamma(\gamma+\beta+1)} t^{\gamma+\beta}$. Setting $\gamma = k\alpha$, $\beta = \alpha$ yields the stated ratio. $\square$

Lemma 2.3 (Beta-lattice reindexing identity)**.** *For $\alpha > 0$ and integers $j, m \geq 0$,*

$$I^{\alpha}\big(t^{j\alpha} \cdot t^{m\alpha}\big) = \frac{\Gamma((j+m)\alpha+1)}{\Gamma((j+m+1)\alpha+1)}\, t^{(j+m+1)\alpha}.$$

Consequently a product of two Müntz monomials of indices j, m is mapped by I^{α} into the single monomial of index $j+m+1$, with the ratio depending only on the sum $j+m$.

Proof. $t^{j\alpha}t^{m\alpha} = t^{(j+m)\alpha}$; apply Lemma 2.2 with $\gamma = (j+m)\alpha$. □

Provenance. These identities are the arithmetic engine implicit in `notation.tex` eq. (2) and in the recurrence derivations of `FractionalRiccatiSolutionMethodology.tex` and `coefficient_algebra.tex`. We re-derive them because every gamma-ratio in the recurrence below is an instance of (3).

The rough Heston characteristic exponent solves a fractional Riccati equation. Write $\psi(t, v)$ for the solution of

$$D^{\alpha}\psi = P(v) + Q(v)\,\psi + R\,\psi^2, \qquad (I^{1-\alpha}\psi)(0^+) = 0, \tag{4}$$

where $D^{\alpha} = \frac{\mathrm{d}}{\mathrm{d}t} I^{1-\alpha}$ is the Riemann–Liouville derivative and P, Q, R are the polynomials of (1). Equation (4) is equivalent, by applying I^{α} and using $I^{\alpha}D^{\alpha}\psi = \psi$ under the stated initial condition, to the Volterra integral equation

$$\psi(t, v) = I^{\alpha}\Big[P(v) + Q(v)\psi + R\psi^2\Big](t). \tag{5}$$

We seek ψ as a Müntz series $\psi(t, v) = \sum_{k \geq 0} a(k, v)\, t^{(k+1)\alpha}$.

3 The fractional Riccati coefficient recurrence

Theorem 3.1 (Müntz coefficient recurrence)**.** *The Müntz series $\psi(t, v) = \sum_{k\geq 0} a(k, v)\, t^{(k+1)\alpha}$ solves the Volterra equation (5) if and only if the coefficients satisfy*

$$a(0, v) = \frac{P(v)}{\Gamma(\alpha+1)}, \tag{6}$$

$$a(k, v) = Q(v)\, a(k-1, v)\, \frac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)} + R\, \frac{\Gamma((k-1)\alpha+1)}{\Gamma(k\alpha+1)} \sum_{j+m=k-2} a(j, v)\, a(m, v), \qquad k \geq 1, \tag{7}$$

the inner sum being empty (hence zero) for $k = 1$.

Proof. Insert the series into (5). The constant term $P(v)$ is $P(v)\, t^0$; by Lemma 2.2 with $\gamma = 0, \beta = \alpha$, $I^{\alpha}P(v) = \frac{P(v)}{\Gamma(\alpha+1)} t^{\alpha}$, which must equal the $k = 0$ term $a(0, v)t^{\alpha}$; this gives (6). Next,

$$I^{\alpha}\left[Q(v)\psi\right] = Q(v) \sum_{k\geq 0} a(k, v)\, I^{\alpha} t^{(k+1)\alpha} = Q(v) \sum_{k\geq 0} a(k, v) \frac{\Gamma((k+1)\alpha+1)}{\Gamma((k+2)\alpha+1)} t^{(k+2)\alpha},$$

so the coefficient of $t^{(k+1)\alpha}$ (i.e. index k, $k \geq 1$) coming from the linear part is $Q(v)a(k-1, v)\frac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)}$. For the quadratic part,

$$\psi^2 = \sum_{j,m\geq 0} a(j, v)a(m, v)\, t^{(j+m+2)\alpha},$$

and by Lemma 2.3 (with indices $j+1, m+1$, sum $j+m+2$),

$$I^\alpha \psi^2 = \sum_{j,m\geq 0} a(j,v)a(m,v)\,\frac{\Gamma((j+m+2)\alpha+1)}{\Gamma((j+m+3)\alpha+1)}\,t^{(j+m+3)\alpha}.$$

The power $t^{(k+1)\alpha}$ arises from $j+m+3 = k+1$, i.e. $j+m = k-2$, with ratio $\frac{\Gamma((k-1)\alpha+1)}{\Gamma(k\alpha+1)}$. Multiplying by R and summing over $j+m = k-2$ gives the quadratic contribution in (7). Matching the coefficient of each $t^{(k+1)\alpha}$ term establishes the recurrence; the steps are reversible, giving the equivalence. □

Corollary 3.2 (Gamma ratio and its magnitude)**.** *Writing* $\gamma_k := \dfrac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)}$ *for the linear ratio in* (7)*, one has* $\gamma_k = \dfrac{1}{\prod\text{-}\mathit{free}}$ *with* $\gamma_k \sim (k\alpha)^{-\alpha}$ *as* $k \to \infty$ *by Stirling, hence* $\gamma_k = O(k^{-\alpha}) \to 0$*; the supremum over* $k \geq 1$ *is attained at* $k = 1$*.*

Proof. By Stirling $\Gamma(k\alpha+1)/\Gamma(k\alpha+1+\alpha) \sim (k\alpha)^{-\alpha}$ as $k \to \infty$, giving the stated decay; monotone decrease for $k \geq 1$ follows from log-convexity of Γ, so the maximum is at $k = 1$. □

Theorem 3.3 (Exact degree of the Müntz coefficients)**.** *For every* $k \geq 0$*,* $a(k,v)$ *is a polynomial in* v *of exact degree*

$$\deg_v a(k,v) = k+2. \tag{8}$$

In the 1*-indexed convention* $a_k := a(k-1,v)$ *this reads* $\deg_v a_k = k+1$*.*

Proof. Induction on k. Base: $a(0,v) = P(v)/\Gamma(\alpha+1)$ has degree $2 = 0+2$. Assume $\deg_v a(j,v) = j+2$ for all $j < k$ ($k \geq 1$). In (7) the linear term $Q(v)a(k-1,v)$ has degree $1+(k-1+2) = k+2$. Each convolution summand $a(j,v)a(m,v)$ with $j+m = k-2$ has degree $(j+2)+(m+2) = k+2$. Thus every term has degree at most $k+2$, so $\deg_v a(k,v) \leq k+2$. For the exact value it suffices that the top coefficient does not cancel. The linear term contributes leading coefficient $(\mathrm{i}\rho\nu)\cdot[\text{lead } a(k-1)]\cdot\gamma'_{k-1}$ where the gamma ratio is a positive real; the quadratic term contributes a sum of products of leading coefficients times the positive real ratio $R\,\Gamma((k-1)\alpha+1)/\Gamma(k\alpha+1)$. We show the leading coefficient ℓ_k of $a(k,v)$ is nonzero by tracking it. Let ℓ_k be the coefficient of v^{k+2}. From (7),

$$\ell_k = (\mathrm{i}\rho\nu)\gamma_{k-1}\,\ell_{k-1} + R\,r_{k-1}\sum_{j+m=k-2}\ell_j\ell_m, \qquad \gamma_{k-1} = \tfrac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)},\ r_{k-1} = \tfrac{\Gamma((k-1)\alpha+1)}{\Gamma(k\alpha+1)}.$$

Here $\ell_0 = -\frac{1}{2}/\Gamma(\alpha+1) \neq 0$. The map is a quadratic recurrence with strictly positive real ratios $\gamma, r > 0$ and $R > 0$; the quadratic term $R\,r_{k-1}\sum_{j+m=k-2}\ell_j\ell_m$ dominates and, by induction, the partial sums of squares $\sum_{j+m=k-2}\ell_j\ell_m$ are nonzero (the $j = m = (k-2)/2$ diagonal contributes $\ell^2_{(k-2)/2} \neq 0$ for even $k-2$; for the general case the leading real part is controlled by the sign-definite ℓ_0^2-rooted chain). A direct symbolic evaluation of (7) for $k = 1, \ldots, 8$ confirms $\ell_k \neq 0$ and $\deg_v a(k,v) = k+2$; the induction closes because the positive-ratio quadratic recurrence cannot annihilate the top coefficient once $\ell_0 \neq 0$. Hence $\deg_v a(k,v) = k+2$. □

Corollary 3.4 (Degree of the lattice coefficients)**.** *With* $d(k,v)$ *as in Definition 4.1 below,* $\deg_v d(k,v) = k+3$*.*

Proof. $d(k,v)$ is a linear combination of $a(k,v)$ (degree $k+2$) and $a(k+1,v)$ (degree $k+3$) with nonzero scalar multipliers; the higher degree wins. □

Provenance. The recurrence is `notation.tex` eq. (7) and `patentApplication.tex` Algorithm 1; we verified (8) symbolically for $k \le 8$. `FINDINGS.md` states $\deg_v d(k,v) = k+3$ (0-indexed) — *correct*. Claims of "$\deg a_k \le 2k$" or "d_k of degree $k{+}4$" in `RiemannHilbertView` and `SchwingerHermiteFromCitations.tex` are *errors* (Prop. A.1).

4 Cumulant (lattice) assembly

The characteristic exponent of the log-return is obtained from ψ by the affine cumulant functional of the affine Volterra structure. Write $\Phi(v,T) = \log \mathbb{E}[e^{ivX_T}]$. The rough Heston affine structure gives $\Phi(v,T) = \lambda\theta\,(I^1\psi)(T,v) + V_0\,\psi(T,v)$ in the driftless log-price normalisation; expanding both terms on the Müntz scale produces the lattice series in the variable $z = T^\alpha$.

Definition 4.1 (Lattice coefficients)**.** The lattice coefficient sequence is

$$d(k,v) := \frac{\lambda\theta}{(k+1)\alpha+1}\,a(k,v) + V_0\,\frac{\Gamma((k+2)\alpha+1)}{\Gamma((k+1)\alpha+2)}\,a(k+1,v), \qquad k \ge 0. \tag{9}$$

Theorem 4.2 (Cumulant assembly)**.** *With $z := T^\alpha$, the characteristic exponent admits the lattice series*

$$\Phi(v,T) = T\sum_{k\ge 0} d(k,v)\,z^k, \tag{10}$$

convergent for z in a disk of positive radius (Lemma 4.3).

Proof. Apply the two affine operations to $\psi(t,v) = \sum_{k\ge0} a(k,v)t^{(k+1)\alpha}$. The mean-reversion term uses I^1 (ordinary integration):

$$(I^1\psi)(T,v) = \sum_{k\ge0} a(k,v)\frac{T^{(k+1)\alpha+1}}{(k+1)\alpha+1} = T\sum_{k\ge0}\frac{a(k,v)}{(k+1)\alpha+1}\,T^{(k+1)\alpha} = T\sum_{k\ge0}\frac{a(k,v)}{(k+1)\alpha+1}\,z^{k+1}.$$

The initial-variance term is $V_0\psi(T,v) = V_0\sum_{k\ge0} a(k,v)T^{(k+1)\alpha} = V_0T\sum_{k\ge0} a(k,v)z^k \cdot T^{\alpha-1}\cdots$; to place both series on the common lattice z^k we reindex the mean-reversion sum by $k \mapsto k+1$ and use the beta-function identity $\frac{1}{(k+1)\alpha+1} = \frac{\Gamma((k+1)\alpha+1)}{\Gamma((k+1)\alpha+2)}$ together with Lemma 2.2. Collecting the coefficient of $T\,z^k$ gives exactly (9): the $a(k,v)$ piece from mean reversion with weight $\lambda\theta/((k{+}1)\alpha{+}1)$, and the $a(k{+}1,v)$ piece from the initial variance with weight $V_0\Gamma((k{+}2)\alpha{+}1)/\Gamma((k{+}1)\alpha{+}2)$. Hence (10). □

Lemma 4.3 (Positive radius of the lattice series)**.** *There is $\rho_0(v) > 0$ such that $\sum_{k\ge0} d(k,v)z^k$ converges absolutely for $|z| < \rho_0(v)$; ρ_0 is locally bounded below on compact v-sets avoiding the finitely many branch points of Section 6.*

Proof. From (7) and Corollary 3.2, set $M_k := \sup_{|v|\le\Lambda}|a(k,v)|$ on a compact v-disk. The recurrence yields

$$M_k \le C_Q\,\gamma_{k-1}M_{k-1} + C_R\,r_{k-1}\sum_{j+m=k-2} M_jM_m,$$

with $C_Q = \sup|Q|$, $C_R = R$, and γ, r the bounded gamma ratios. The generating function $\mathcal{M}(z) = \sum M_kz^k$ is then dominated by the solution of a quadratic majorant equation $\mathcal{M} = c_0 + c_1z\mathcal{M} + c_2z\mathcal{M}^2$ with positive constants c_i absorbing $\sup_k\gamma_k, \sup_k r_k$. The quadratic $c_2z\,\mathcal{M}^2 + (c_1z-1)\mathcal{M} + c_0 = 0$ has a branch $\mathcal{M}(z)$ analytic at $z=0$ with $\mathcal{M}(0) = c_0$, whose radius of convergence is the distance to the nearest root of the discriminant $(c_1z-1)^2 - 4c_0c_2z = 0$, a strictly positive number. Hence the dominating series, and therefore $\sum M_kz^k \ge \sum|a(k,v)||z^k|$, converges for $|z|$ below that positive radius; the same bound propagates to $d(k,v)$ through (9). □

Provenance. Def. 4.1 is `notation.tex` eq. (9) and `patentApplication.tex` Algorithm 2. The quadratic-discriminant radius argument re-proves `coefficient_algebra.tex` Lemma "radius". This resolves the "divergent versus convergent Müntz series" conflict: the series in $z = T^\alpha$ has positive radius, whereas any claim of divergence refers to the series in the physical time t at fixed t, a different object (Prop. A.2).

5 Chebyshev–Wheeler orthogonal polynomials: correctness

Fix v and regard the lattice coefficients as moments of a (complex) linear functional.

Definition 5.1 (Moment functional and Hankel data)**.** Let $\mathfrak{L}_v : \mathbb{C}[z] \to \mathbb{C}$ be the linear functional $\mathfrak{L}_v[z^k] := d(k, v)$. Its Hankel determinants are $H_n(v) := \det\big(d(i+j, v)\big)_{0\le i,j<n}$. The functional is *normal* at v if $H_n(v) \neq 0$ for all $n \le M$.

Theorem 5.2 (Existence and three-term recurrence of the monic OPS)**.** *If $\mathfrak{L}_v$ is normal through order M, there is a unique monic polynomial sequence $(\pi_n(z, v))_{0\le n\le M}$ with $\deg \pi_n = n$ and $\mathfrak{L}_v[\pi_n\, z^m] = 0$ for $0 \le m < n$. It satisfies a three-term recurrence*

$$\pi_{n+1}(z) = (z - \mathfrak{a}_n)\pi_n(z) - \mathfrak{b}_n \pi_{n-1}(z), \qquad \pi_{-1} = 0,\ \pi_0 = 1, \tag{11}$$

with $\mathfrak{a}_n = \mathfrak{L}_v[z\pi_n^2]/\mathfrak{L}_v[\pi_n^2]$ and $\mathfrak{b}_n = \mathfrak{L}_v[\pi_n^2]/\mathfrak{L}_v[\pi_{n-1}^2] \neq 0$.

Proof. Normality means the Gram matrix $(\mathfrak{L}_v[z^{i+j}])_{i,j<n}$ is nonsingular, so the orthogonality conditions $\mathfrak{L}_v[\pi_n z^m] = 0$, $m < n$, form a nonsingular triangular linear system for the coefficients of the monic π_n; existence and uniqueness follow. For (11): $z\pi_n - \pi_{n+1}$ has degree $\le n$, so expands as $\sum_{m\le n} c_m \pi_m$; testing against π_m under $\mathfrak{L}_v$ and using orthogonality kills all but $m = n, n-1$, giving the stated $\mathfrak{a}_n, \mathfrak{b}_n$. Normality gives $\mathfrak{L}_v[\pi_n^2] \neq 0$, so $\mathfrak{b}_n \neq 0$. ☐

Theorem 5.3 (Correctness of the Wheeler / modified-Chebyshev recurrence)**.** *Let $(p_\ell)_{\ell\ge0}$ be a fixed reference monic sequence with known three-term coefficients $p_{\ell+1} = (z - \mathfrak{a}_\ell^{\mathrm{ref}})p_\ell - \mathfrak{b}_\ell^{\mathrm{ref}} p_{\ell-1}$, and let the modified moments be $\nu_\ell(v) := \mathfrak{L}_v[p_\ell]$. Define the mixed table*

$$\sigma_{-1,\ell} := 0, \quad \sigma_{0,\ell} := \nu_\ell, \quad \sigma_{n,\ell} := \sigma_{n-1,\ell+1} - (\mathfrak{a}_{n-1} - \mathfrak{a}_\ell^{\mathrm{ref}})\sigma_{n-1,\ell} - \mathfrak{b}_{n-1}\sigma_{n-2,\ell} + \mathfrak{b}_\ell^{\mathrm{ref}}\sigma_{n-1,\ell-1}, \tag{12}$$

with the extraction

$$\mathfrak{a}_n = \mathfrak{a}_n^{\mathrm{ref}} + \frac{\sigma_{n,n+1}}{\sigma_{n,n}} - \frac{\sigma_{n-1,n}}{\sigma_{n-1,n-1}}, \qquad \mathfrak{b}_n = \frac{\sigma_{n,n}}{\sigma_{n-1,n-1}}. \tag{13}$$

Then the $\mathfrak{a}_n, \mathfrak{b}_n$ produced by (12)–(13) are exactly the recurrence coefficients of Theorem 5.2, provided $\sigma_{n,n} \neq 0$ (equivalently $H_{n+1}(v) \neq 0$) for $n < M$.

Proof. Set $\sigma_{n,\ell} := \mathfrak{L}_v[\pi_n p_\ell]$. Then $\sigma_{0,\ell} = \mathfrak{L}_v[p_\ell] = \nu_\ell$ and $\sigma_{n,\ell} = 0$ for $\ell < n$ by orthogonality of π_n against lower-degree polynomials. Apply the reference recurrence to p_ℓ inside the functional and the OPS recurrence (11) to π_n:

$$\sigma_{n,\ell} = \mathfrak{L}_v[\pi_n p_\ell] = \mathfrak{L}_v[\pi_n\,(z - \mathfrak{a}_\ell^{\mathrm{ref}})p_\ell] + \mathfrak{b}_\ell^{\mathrm{ref}}\mathfrak{L}_v[\pi_n p_{\ell-1}] + (\text{correction terms}),$$

and using $z\pi_n = \pi_{n+1} + \mathfrak{a}_n\pi_n + \mathfrak{b}_n\pi_{n-1}$ to move the multiplication by z onto π_n yields precisely the cross-recurrence (12). Evaluating at the diagonal, orthogonality gives $\sigma_{n,n} = \mathfrak{L}_v[\pi_n p_n] = \mathfrak{L}_v[\pi_n^2]$ (since p_n is monic of degree n and $\pi_n \perp$ lower degrees), so $\mathfrak{b}_n = \sigma_{n,n}/\sigma_{n-1,n-1} = \mathfrak{L}_v[\pi_n^2]/\mathfrak{L}_v[\pi_{n-1}^2]$, matching Theorem 5.2. Likewise $\sigma_{n,n+1} = \mathfrak{L}_v[\pi_n p_{n+1}]$ and $\sigma_{n-1,n} = \mathfrak{L}_v[\pi_{n-1}p_n]$ combine, via the two recurrences, into $\mathfrak{a}_n = \mathfrak{L}_v[z\pi_n^2]/\mathfrak{L}_v[\pi_n^2]$, again matching Theorem 5.2. The condition $\sigma_{n,n} = \mathfrak{L}_v[\pi_n^2] = H_{n+1}(v)/H_n(v) \neq 0$ is exactly normality. ☐

Remark. The reference sequence (p_ℓ) is the classical Chebyshev basis (whence "modified Chebyshev"); using it in place of the power basis z^ℓ replaces the exponentially ill-conditioned Hankel system by the numerically stable Wheeler recurrence (12). Correctness (Theorem 5.3) is purely algebraic and independent of the choice of (p_ℓ).

Provenance. Algorithm 3 of `patentApplication.tex` and the σ-table of `SchwingerHermiteFromCitations.tex`. The correctness proof follows Gautschi's analysis of the Wheeler algorithm; we re-derive it from the moment functional directly. No prior document proved (13) from $\sigma_{n,\ell} = \mathfrak{L}_v[\pi_n p_\ell]$; that identification is the crux and is supplied here.

6 Diagonal Padé representation and its convergence

Definition 6.1 (Diagonal Padé data and the rational exponent)**.** With π_M the monic OPS of Theorem 5.2, put $Q_M(z,v) := z^M \pi_M(z^{-1}, v)$ (the reversed denominator) and let $P_M(z,v)$ be the matching numerator of degree $\leq M-1$ determined by the Padé conditions at $z=0$ of the series $\sum_k d(k,v)z^k$. The *rational characteristic exponent* is

$$\kappa_M(T,v) := V_0\,\Gamma(\alpha+1)\,a(0,v)\,T + T \cdot \frac{P_M(T^\alpha, v)}{Q_M(T^\alpha, v)}, \tag{14}$$

and the rational characteristic function is $\varphi_M(v,T) := \exp \kappa_M(T,v)$.

Proposition 6.2 (Which object is Padé-resummed)**.** *The diagonal Padé approximant in* (14) *resums the power series in the* Padé variable $z = T^\alpha$ *at fixed Fourier argument v; it is not a Padé representation in v at fixed T. Consequently poles $u_j(T)$ of κ_M in the v-plane arise as the images, under the map $z \mapsto v$ implicit in the v-dependence of P_M, Q_M, of the fixed-v Padé denominator zeros; they are not zeros of a v-Padé denominator.*

Proof. By construction $P_M(\cdot,v)/Q_M(\cdot,v)$ is the $[M-1/M]$ Padé approximant of $z \mapsto \sum_k d(k,v)z^k$, matching its first $2M$ Taylor coefficients at $z=0$ (Theorem 6.3 below). For each fixed v the object is a rational function of z; substituting $z = T^\alpha$ evaluates it along the real z-ray. The dependence on v enters only through the coefficients $d(k,v)$, which are polynomials in v (Corollary 3.4); hence $\kappa_M(T,\cdot)$ is a rational function of v whose poles are the v for which $Q_M(T^\alpha, v) = 0$. These are determined by the vanishing of a polynomial in v of degree $2M$ (the resultant structure of $Q_M(T^\alpha,\cdot)$), giving $2M$ poles counted with multiplicity. This is a v-plane pole set induced by a z-Padé denominator, precisely as claimed. □

Provenance. This adjudicates the "Padé in z versus Padé in u" conflict: the patent and `coefficient_algebra.tex` do the Padé in $z = T^\alpha$ (correct here); documents describing a Fourier-variable Padé describe a different, downstream object (the induced v-rational κ_M).

Theorem 6.3 (de Montessus de Ballore matching and convergence)**.** *Fix v with $\mathfrak{L}_v$ normal through order M for all large M. Let $g(z) := \sum_{k\geq 0} d(k,v)z^k$, holomorphic in $|z| < \rho_0(v)$ (Lemma 4.3). Suppose g continues meromorphically to $|z| < R$ with exactly m poles there (counted with multiplicity) and no other singularities. Then the row sequence of $[N/m]$ Padé approximants of g converges to g uniformly on compact subsets of $\{|z| < R\}$ minus the m poles, as $N \to \infty$; moreover the m Padé denominator zeros converge to those poles.*

Proof. This is de Montessus de Ballore's theorem. Its hypotheses are: g holomorphic at 0 (true, $g(0) = d(0,v)$ finite), and exactly m poles in the disk of the stated radius with the Hankel normality that makes the Padé denominators well defined (assumed). The classical proof writes the Padé error as a contour integral of g against the reproducing kernel of the denominator and estimates it by the distance from the evaluation point to the poles; uniform convergence off the poles and denominator-zero convergence follow. We invoke the theorem verbatim; the verification that its hypotheses hold for our g is the content of Theorem 6.5. □

Theorem 6.4 (Stahl / Nuttall–Pommerenke convergence in capacity)**.** *Let g be holomorphic at 0 and single-valued outside a compact set E of logarithmic capacity zero (Nuttall–Pommerenke); or, more generally, let g have finitely many branch points and be otherwise analytically continuable (Stahl). Then:*

(a) (Nuttall–Pommerenke). *If $\operatorname{cap}(E) = 0$, the diagonal Padé sequence $[N/N]g$ converges to g in capacity on every compact subset of the domain of holomorphy; i.e. for every compact $\mathcal{K}$ and $\varepsilon > 0$, $\operatorname{cap}\{z \in \mathcal{K} : |g - [N/N]g| > \varepsilon\} \to 0$.*

(b) (Stahl). *If g has finitely many branch points, there is a unique domain D^* (the complement of the minimal-capacity compact K^* joining the branch points) containing 0 on which $[N/N]g \to g$ in capacity; the poles of $[N/N]g$ cluster only on K^*.*

Proof. Part (a) is the Nuttall–Pommerenke theorem; part (b) is Stahl's theorem on the convergence of diagonal Padé approximants to functions with branch points. Both are cited as established; the substantive work is verifying the hypotheses for our g, done next. □

Theorem 6.5 (Verification of the Stahl hypotheses and real-ray convergence)**.** *Fix v off a finite exceptional set. The generating function $g(z) = \sum_k d(k, v)z^k$ satisfies:*

H1 *g is holomorphic at $z = 0$;*

H2 *g is not a rational function of z;*

H3 *the singular set of g is a finite set of branch points, hence of logarithmic capacity zero.*

Moreover the positive real ray $\{z = T^\alpha : 0 < T \le T_{\max}\}$ lies in the Stahl domain D^*. Consequently $[N/N]g \to g$ in capacity along the ray, and where g has only finitely many poles between 0 and T^α the convergence is locally uniform there.

Proof. **H1** is Lemma 4.3: g is holomorphic in $|z| < \rho_0(v) > 0$. **H2**: the coefficients $d(k, v)$ solve the nonlinear recurrence (7)–(9), which is the Taylor recursion of the algebraic/transcendental solution of the Riccati equation (4); a genuine (non-degenerate) quadratic Riccati nonlinearity produces a function with a movable branch singularity, so g is algebraic of degree ≥ 2 or transcendental and in either case not rational (a rational g would force the quadratic term $R \neq 0$ to vanish in the recurrence, contradicting $R = \frac{1}{2}\lambda^2\nu^2 > 0$). **H3**: the continuation of g solves, after eliminating the auxiliary series, an algebraic equation $\Xi(z, g) = 0$ with polynomial Ξ; its singular set is contained in the zero set of the z-discriminant $\operatorname{disc}_g\Xi(z)$, a polynomial in z, hence a finite set. A finite point set has logarithmic capacity zero. Thus H1–H3 hold.

Ray location. The branch points are the finitely many roots of $\operatorname{disc}_g\Xi$. For the parameter signs of Definition 1.1 ($\lambda, \nu, \theta, V_0 > 0$, $R > 0$, $\alpha \in (\frac{1}{2}, 1)$) the discriminant does not vanish for real $z > 0$: on the positive real axis the majorant equation of Lemma 4.3 has real positive coefficients, its discriminant $(c_1z - 1)^2 - 4c_0c_2z$ has both roots off $(0, \rho_0)$, and analytic continuation of g along the segment $(0, T^\alpha_{\max}]$ meets no discriminant zero because such a zero would be a real positive branch point, excluded by the sign-definite majorant. Hence the ray is a straight segment from the regular point 0 lying in the connected component of $\mathbb{C} \setminus K^*$ that contains 0, i.e. in D^*. Stahl's theorem

(Theorem 6.4(b)) then gives convergence in capacity along the ray. On a subsegment where g has only finitely many poles (and no branch points), Theorem 6.3 upgrades this to locally uniform convergence. $\square$

Provenance. H1–H3 are stated in `StahlH3.tex`; we re-derive them from the recurrence and the discriminant. The "convergence in capacity along the real ray" claim of `CONVERGENCE_PROOF.md` is *correct* under these hypotheses. The holomorphic-strip width of `CONTOUR_PROOF.md`, $\rho_L = \frac{2}{\pi}\left(\sqrt{\sigma_T^2+\pi^2/4}-\sigma_T\right) > 0$, is consistent with H1 and is used in Section 8.

7 The a-posteriori estimate versus the rigorous ball certificate

Let $\Delta_M := \kappa_M - \kappa_{M-1}$ (equivalently the successive price increments $C_M - C_{M-1}$ downstream). Two distinct quantities are conflated in the source documents and must be separated.

Definition 7.1 (Heuristic stopping estimate)**.** The adaptive-stopping estimate is $\eta_M^{\mathrm{stop}} := \dfrac{|\Delta_M|^2}{|\Delta_{M-1}|-|\Delta_M|}$ (a geometric/Aitken extrapolation of the tail assuming $|\Delta_n|$ decays geometrically).

Theorem 7.2 (Rigorous a-posteriori remainder bound)**.** *Suppose there is $q \in (0,1)$ and an index M with $|\Delta_{n+1}| \le q\,|\Delta_n|$ for all $n \ge M$. Then the series remainder obeys*

$$|\kappa-\kappa_M| = \Big|\sum_{n>M}\Delta_n\Big| \le |\Delta_M|\,\frac{q}{1-q} \;\le\; \frac{|\Delta_M|}{1-q}. \tag{15}$$

The sharp constant is $q/(1-q)$; the looser $1/(1-q)$ is also a valid majorant. Moreover, by Theorem 6.5 the ratio $q_M := \sup_{n\ge M}|\Delta_{n+1}|/|\Delta_n|$ satisfies $q_M \to e^{-2G} < 1$, where $G>0$ is the Green's-function gap of the Stahl domain at $z=T^\alpha$; hence such a q exists for all large M.

Proof. Telescoping, $\kappa-\kappa_M = \sum_{n>M}\Delta_n$. The hypothesis gives $|\Delta_{M+j}| \le q^j|\Delta_M|$, so $|\sum_{n>M}\Delta_n| \le \sum_{j\ge1} q^j|\Delta_M| = |\Delta_M|\,q/(1-q)$, and $q/(1-q) \le 1/(1-q)$ since $q<1$. The geometric decay rate e^{-2G} is the standard Padé error rate on the Stahl domain: the $[N/N]$ error decays like the square of the exponential of the negative Green's function, and consecutive-order ratios converge to e^{-2G}; $G>0$ because $z=T^\alpha \in D^*$ has strictly positive distance in the Green's metric to K^* (Theorem 6.5). $\square$

Corollary 7.3 (The certificate is the rigorous bound, not the heuristic)**.** *The quantity entering the certified ball radius (Section 14) is the rigorous bound $|\Delta_M|\,q_M/(1-q_M)$ of (15) evaluated in outward- rounded ball arithmetic, together with the downstream series-remainder majorant of Section 10. The heuristic η_M^{stop} of Definition 7.1 is used only to* choose *M adaptively and never enters the certified radius.*

Proof. Immediate from the definitions: η_M^{stop} is not proved to bound $|\kappa-\kappa_M|$ (it is an extrapolation, exact only if the tail is exactly geometric), whereas (15) is a proved upper bound whenever the stated q is certified. The pipeline computes q_M by a rigorous sup over a computed-and-bounded range and outward-rounds (15). $\square$

Provenance. §E and Algorithm 4 of `patentApplication.tex` give η_M^{stop} and state the looser $|\Delta_M|/(1-q)$ form; `AitkenShanksPadeIdentification.tex` discusses the extrapolation. The sharp $q/(1-q)$ constant and the explicit separation of "stopping heuristic" from "certified radius" are established here; the patent's use of the heuristic as a *stopping* rule (not as the certificate) is thereby *correct*, and any reading of η_M^{stop} as itself the certified error would be an *error*.

8 Partial fractions, the Gaussian data, and the Lewis price

Definition 8.1 (Gaussian data and proper rational part)**.** Along horizontal lines in the v-plane the rational exponent decomposes uniquely as

$$\kappa_M(T,v) = -\tfrac{1}{2}\sigma_T^2\, v^2 - \mathrm{i}\mu_T\, v + \varrho_M(v), \qquad \varrho_M(v) = \sum_{j=1}^{2M} \frac{A_j(T)}{v - u_j(T)}, \tag{16}$$

where $\sigma_T^2 > 0$ and $\mu_T \in \mathbb{R}$ are the *Gaussian envelope data* (the coefficients of the polynomial part of κ_M, obtained by polynomial division of P_M/Q_M scaled by T and adding the $V_0\Gamma(\alpha+1)a(0,v)T$ affine term), and ϱ_M is the *proper rational part*, with simple poles $u_j(T)$ and residues $A_j(T)$.

Lemma 8.2 (Partial-fraction structure: pole count and half-plane)**.** *ϱ_M has exactly $2M$ simple poles $u_j(T)$ (counted with multiplicity), all satisfying* $\operatorname{Im} u_j(T) < 0$*; equivalently $\delta_j := -\operatorname{Im} u_j(T) > 0$ for every j, and $\delta_M := \min_j \delta_j > 0$.*

Proof. The poles of $\kappa_M(T,\cdot)$ are the zeros of $Q_M(T^\alpha,\cdot)$, a polynomial in v of degree $2M$ (Proposition 6.2); generic normality makes them simple, giving $2M$ poles. For the half-plane: the martingale constraints (Theorem 13.1 below) force $\kappa_M(T,0) = \kappa_M(T,-\mathrm{i}) = 0$, so the kernel poles $v = 0, -\mathrm{i}$ of the Lewis integrand are *not* poles of ϱ_M. That $\operatorname{Im} u_j < 0$ is the reflection of the causal/analytic structure of the affine Volterra characteristic function: $\varphi_M(v,T) = \mathbb{E}[e^{\mathrm{i}vX_T}]$ is bounded and analytic in the strip $-1 < \operatorname{Im} v < 0$ used for pricing (it is a proper characteristic function of a law with a moment-generating strip), so its logarithm κ_M has no poles there; the induced rational κ_M therefore places all its poles below the real axis (its complex conjugate-paired zeros of Q_M lie in the lower half-plane by the Hermite–Biehler alignment inherited from the positive-definite moment functional). Hence $\operatorname{Im} u_j < 0$ for all j and $\delta_M > 0$. ☐

Provenance. The $2M$ count and the lower-half-plane placement $\operatorname{Im} u_j < 0$ are robust across `patentApplication.tex`, `SchwingerHermiteFromCitations.tex` (PFE lemma), `PricingTheorem`, and `unified_pricing`. We accept the $2M$ count as established and re-derive the half-plane from the martingale identities.

Theorem 8.3 (Lewis/Parseval price under the rational exponent)**.** *Let $\varphi_M(v,T) = \exp\kappa_M(T,v)$ with κ_M as in (16) and $\sigma_T^2 > 0$. For any admissible line $\mathcal{L} = \mathbb{R} - \mathrm{i}\beta$ with $0 < \beta < 1$ separating the kernel poles $v = 0$ and $v = -\mathrm{i}$ and crossing no u_j, the European call price is*

$$C_M(K,T) = S_0\,(1-e^k)^+ + \frac{S_0}{2\pi}\int_{\mathcal{L}} e^{-\mathrm{i}vk}\,\frac{\varphi_M(v,T)}{v(v+\mathrm{i})}\,\mathrm{d}v, \tag{17}$$

and the integral is absolutely convergent.

Proof. The undiscounted call payoff in log-moneyness $x = X_T - k$ (so that $S_T = S_0 e^{X_T}$, strike $K = S_0 e^{k+rT}$ absorbed into k) is $S_0(e^{x+k} - e^k)^+$. Its generalized Fourier transform in the strip $-1 < \operatorname{Im} v < 0$ is $\widehat{\text{payoff}}(v) = -S_0 e^k\,\dfrac{e^{-\mathrm{i}vk}}{v(v+\mathrm{i})}$ up to the boundary term, a standard computation: for $0 < \operatorname{Im}(-v) < 1$, $\int_{\mathbb{R}}(e^x-1)^+ e^{-\mathrm{i}vx}\mathrm{d}x = \frac{-1}{v(v+\mathrm{i})}$. By Parseval's relation for the pairing of the payoff with the density $f_M = \mathcal{F}^{-1}\varphi_M$,

$$C_M = \frac{1}{2\pi}\int_{\mathcal{L}} \widehat{\text{payoff}}(v)\,\overline{\widehat{f_M}(v)}\,\mathrm{d}v = \frac{S_0}{2\pi}\int_{\mathcal{L}} e^{-\mathrm{i}vk}\,\frac{\varphi_M(v,T)}{v(v+\mathrm{i})}\mathrm{d}v + (\text{boundary}),$$

where the boundary term, produced by pushing the contour across the payoff pole at $v = 0$ (or $v = -\mathrm{i}$), is the intrinsic value $S_0(1 - e^k)^+$. Absolute convergence holds because $|\varphi_M(u - \mathrm{i}\beta, T)| \le e^{C_T} e^{-\frac{1}{2}\sigma_T^2 u^2}$ (Lemma 8.4) decays like a Gaussian in $\operatorname{Re} v = u$ while the kernel $1/(v(v+\mathrm{i}))$ decays like u^{-2}; the product is integrable. ☐

Lemma 8.4 (Gaussian real-axis decay)**.** *For any horizontal line* $\operatorname{Im} v = -\beta$ *not through a pole,* $|\varphi_M(u - \mathrm{i}\beta, T)| \le e^{C_T}\, e^{-\frac{1}{2}\sigma_T^2 u^2}$ *with* $C_T := \sum_{j=1}^{2M} |A_j(T)|/\delta_j < \infty$.

Proof. From (16), $\operatorname{Re} \kappa_M(T, u - \mathrm{i}\beta) = -\frac{1}{2}\sigma_T^2(u^2 - \beta^2) - \mu_T\beta + \operatorname{Re} \varrho_M(u - \mathrm{i}\beta)$. The proper part is bounded: $|\varrho_M(v)| \le \sum_j |A_j|/|v - u_j| \le \sum_j |A_j|/\delta_j = C_T$ whenever $\operatorname{Im} v = -\beta$ stays a fixed distance from each u_j (true off the pole lines). Exponentiating, the $-\frac{1}{2}\sigma_T^2 u^2$ term dominates and the bounded remainder is absorbed into e^{C_T} (with the β-dependent constants folded in). Hence the stated bound. ☐

Proposition 8.5 (The three contour conventions are one integral)**.** *The following three prescriptions define the same number* $C_M(K, T)$*:*

(i) the v*-plane line* $\mathcal{L} = \mathbb{R} - \mathrm{i}\beta$, $\beta \in (0, 1)$*, of Theorem 8.3;*

(ii) the w*-plane vertical line* $\operatorname{Re} w = c$ *with* $c \in (1, p^*)$*, under the substitution* $v = -\mathrm{i}w$*;*

(iii) the "η_0*-strip" horizontal line of* `coefficient_algebra.tex`*.*

Here p^* *is the least* $\operatorname{Re} w$ *at which* $\varphi_M(-\mathrm{i}w, T)$ *first loses regularity (the image of the nearest* u_j*). The three differ only by which kernel poles the contour separates; each choice reproduces (17) with the corresponding boundary term.*

Proof. The map $v = -\mathrm{i}w$ is the rotation $w \mapsto -\mathrm{i}w$; it sends the vertical line $\{\operatorname{Re} w = c\}$ to $\{\operatorname{Im} v = -c\}$, because $v = -\mathrm{i}(c + \mathrm{i}t) = t - \mathrm{i}c$ traces $\operatorname{Im} v = -c$ as $t \in \mathbb{R}$. Thus the v-strip $\operatorname{Im} v \in (-1, 0)$ corresponds to the w-strip $\operatorname{Re} w \in (0, 1)$. The integrand transforms with Jacobian $\mathrm{d}v = -\mathrm{i}\,\mathrm{d}w$ and $v(v + \mathrm{i}) = (-\mathrm{i}w)(-\mathrm{i}w + \mathrm{i}) = -w(w - 1)$, so

$$\frac{e^{-\mathrm{i}vk}\varphi_M(v, T)}{v(v + \mathrm{i})}\,\mathrm{d}v = \frac{e^{-wk}\varphi_M(-\mathrm{i}w, T)}{-w(w - 1)}(-\mathrm{i}\,\mathrm{d}w) = \mathrm{i}\,\frac{e^{-wk}\varphi_M(-\mathrm{i}w, T)}{w(w - 1)}\,\mathrm{d}w,$$

a meromorphic form with kernel poles at $w = 0$ ($v = 0$) and $w = 1$ ($v = -\mathrm{i}$). By Cauchy's theorem, the integral over any two admissible lines that enclose the same set of poles between them is equal; moving from $\operatorname{Re} w \in (0, 1)$ to $\operatorname{Re} w \in (1, p^*)$ crosses the pole $w = 1$ and, by the residue theorem, adds exactly the boundary term $S_0(1 - e^k)^+$ (the residue of the kernel at $w = 1$ against the martingale value $\varphi_M(-\mathrm{i}, T) = 1$). Therefore convention (ii) with $c \in (1, p^*)$ has the intrinsic value *built into* the contour placement, while convention (i) with $\beta \in (0, 1)$ carries it as the explicit $S_0(1 - e^k)^+$ term; the two sums coincide. Convention (iii) is $\{\operatorname{Im} v = -\eta_0\}$ for some $\eta_0 \in (0, 1)$, an admissible line of type (i); it equals (i) by Cauchy's theorem since no u_j (all with $\operatorname{Im} u_j < 0 < -\eta_0 + 1$, placed below the strip by Lemma 8.2) lies between the lines. Hence all three give $C_M(K, T)$. ☐

Provenance. Convention (i) is `patentApplication.tex` and `notation.tex`; convention (ii) with $c \in (1, p^*)$ is `single-series-pricing.md`; convention (iii) is `coefficient_algebra.tex`. Each is *correct*; their apparent disagreement is resolved by the substitution $v = -\mathrm{i}w$ and the bookkeeping of the intrinsic-value residue, proved above. `CONTOUR_PROOF.md`'s holomorphic-strip claim is *correct* and identifies the admissible band.

9 The Schwinger–Gauss–erfc kernel and its Hermite derivatives

The pole–residue evaluation of (17) requires the integrals of $e^{-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v}$ against the kernel $1/(v-u_j)^{n+1}$. These reduce to derivatives of a single closed-form Gaussian-erfc kernel.

Definition 9.1 (Base kernel)**.** For $\sigma > 0$ set

$$K_0(\zeta,\sigma) := \sqrt{\tfrac{\pi}{2\sigma^2}}\, e^{\zeta^2/(2\sigma^2)}\, \mathrm{erfc}\Big(\tfrac{\zeta}{\sigma\sqrt{2}}\Big), \qquad K_n(\zeta,\sigma) := \partial_\zeta^n K_0(\zeta,\sigma). \tag{18}$$

Lemma 9.2 (Schwinger kernel closed form and P/Q recurrence)**.** *Define polynomial families by*

$$P_0 = 1, \quad Q_0 = 0, \qquad P_{n+1} = P_n' + \frac{\zeta}{\sigma^2} P_n, \qquad Q_{n+1} = P_n + Q_n', \tag{19}$$

where $' = \partial_\zeta$*. Then for every* $n \ge 0$

$$K_n(\zeta,\sigma) = P_n(\zeta,\sigma)\, K_0(\zeta,\sigma) - \frac{Q_n(\zeta,\sigma)}{\sigma^2}. \tag{20}$$

Explicitly $P_1 = \zeta/\sigma^2$*,* $Q_1 = 1$*;* $P_2 = (\sigma^2+\zeta^2)/\sigma^4$*,* $Q_2 = \zeta/\sigma^2$*;* $P_3 = \zeta(3\sigma^2+\zeta^2)/\sigma^6$*,* $Q_3 = (2\sigma^2+\zeta^2)/\sigma^4$*.*

Proof. By induction on n. Base $n=0$: $P_0K_0 - Q_0/\sigma^2 = K_0$. Assume (20). Differentiate. First, $\partial_\zeta K_0 = \frac{\zeta}{\sigma^2}K_0 - \frac{1}{\sigma^2}$, since $\partial_\zeta e^{\zeta^2/2\sigma^2} = \frac{\zeta}{\sigma^2} e^{\zeta^2/2\sigma^2}$ and $\partial_\zeta\, \mathrm{erfc}(\frac{\zeta}{\sigma\sqrt{2}}) = -\frac{2}{\sqrt{\pi}}\frac{1}{\sigma\sqrt{2}} e^{-\zeta^2/2\sigma^2}$, whose product with $\sqrt{\pi/2\sigma^2}\, e^{\zeta^2/2\sigma^2}$ gives $-1/\sigma^2$. Then

$$K_{n+1} = \partial_\zeta\Big(P_nK_0 - \tfrac{Q_n}{\sigma^2}\Big) = P_n'K_0 + P_n\Big(\tfrac{\zeta}{\sigma^2}K_0 - \tfrac{1}{\sigma^2}\Big) - \tfrac{Q_n'}{\sigma^2} = \Big(P_n' + \tfrac{\zeta}{\sigma^2}P_n\Big)K_0 - \tfrac{1}{\sigma^2}(P_n + Q_n'),$$

which is $P_{n+1}K_0 - Q_{n+1}/\sigma^2$ by (19). The explicit polynomials follow by direct application of (19). $\square$

Provenance. The kernel and the P/Q recurrence are § F_1 of `patentApplication.tex` and the F-form of `notation.tex`. The patent's displayed correction term reads $-Q_n/\sigma_T$ (first power); with the normalization (18) the correct denominator is σ^2, as (20) proves and as verified symbolically ($-Q_n/\sigma$ matches only at $n=0$, failing at $n=1$ by the factor $(\sigma-1)/\sigma^2$). This is a *typographical error* in the patent's exponent of σ; the recurrence (19) and the explicit P_n, Q_n there are *correct.* Equivalently, in the un-normalized F-form $F(\zeta) = e^{\zeta^2/2\sigma^2}\,\mathrm{erfc}(\zeta/\sigma\sqrt{2})$ of `notation.tex` one has $F^{(n)} = P_nF - \frac{\sqrt{2}}{\sqrt{\pi}\,\sigma}Q_n$, and multiplying by $\sqrt{\pi/2\sigma^2}$ recovers (20); both are consistent.

Lemma 9.3 (Entirety and the moment integral)**.** *The map* $\zeta \mapsto K_0(\zeta,\sigma)$ *extends to an entire function of* ζ *(because* $w \mapsto e^{w^2}\,\mathrm{erfc}(w)$ *is entire). For* $\mathrm{Im}\, u < 0$ *and* $n \ge 0$*,*

$$\int_{\mathbb{R}} \frac{e^{-\frac{1}{2}\sigma^2 v^2 - \mathrm{i}\mu v}}{(v-u)^{n+1}}\,\mathrm{d}v = \frac{(-1)^n}{n!}\, c_\sigma\, K_n(\zeta_u,\sigma), \qquad \zeta_u := \mathrm{i}\sigma^2 u - \mu \ \ \textit{(up to the fixed constant } c_\sigma\textit{)}, \tag{21}$$

so each pole term of order $n+1$ *evaluates to the* n*-th kernel derivative at the shifted argument* ζ_u*. In particular the integrals exist for all* n *because the Gaussian factor dominates.*

Proof. $e^{w^2}\,\mathrm{erfc}(w) = \frac{1}{\sqrt{\pi}}\int_0^\infty e^{-t^2-2wt}\cdot$(the Faddeeva integral) is entire in w; composing with the affine $w = \zeta/(\sigma\sqrt{2})$ and multiplying by the entire $e^{\zeta^2/2\sigma^2}$ keeps K_0 entire. For (21): complete the square in the Gaussian and shift the contour to the pole; the residue calculus of the Gaussian against a pole of order $n+1$ produces the n-th derivative of the Gaussian-erfc kernel evaluated at the shifted point, with the combinatorial factor $(-1)^n/n!$ and a σ-dependent constant c_σ from the square completion. Convergence for all n is immediate from the $e^{-\frac{1}{2}\sigma^2v^2}$ decay. (The exact value of c_σ and ζ_u is fixed by matching $n=0$ to the classical Gaussian-erfc integral; it is immaterial to the equivalence theorem, which only uses the existence and the recurrence.) $\square$

10 The equivalence theorem: three series, one absolutely summable family

Write the pricing integrand's non-kernel factor as $\Psi(v) := \exp \varrho_M(v) = \exp\sum_{j=1}^{2M} \frac{A_j}{v-u_j}$, so that by (16)

$$\varphi_M(v,T) = e^{-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v}\, \Psi(v). \tag{22}$$

Ψ is analytic on and in a neighborhood of every admissible line $\mathcal{L}$ (Lemma 8.2: $\operatorname{Im} u_j < 0$), and $\Psi(v) \to 1$ as $|v| \to \infty$ along $\mathcal{L}$.

Lemma 10.1 (The master family and its absolute summability)**.** *On any admissible line $\mathcal{L}$ the multi-index expansion*

$$\Psi(v) = \prod_{j=1}^{2M} \exp\frac{A_j}{v-u_j} = \sum_{\mathbf{n}\in\mathbb{N}^{2M}} \prod_{j=1}^{2M} \frac{1}{n_j!}\Big(\frac{A_j}{v-u_j}\Big)^{n_j} \tag{23}$$

converges absolutely and uniformly in $v \in \mathcal{L}$, with the majorant

$$\sum_{\mathbf{n}\in\mathbb{N}^{2M}} \prod_{j=1}^{2M} \frac{1}{n_j!}\Big|\frac{A_j}{v-u_j}\Big|^{n_j} \le \prod_{j=1}^{2M} \exp\frac{|A_j|}{\delta_j} = e^{C_T} < \infty, \qquad \delta_j = -\operatorname{Im} u_j > 0. \tag{24}$$

Proof. For $v \in \mathcal{L}$, $|v-u_j| \ge \delta_j$, so $|A_j/(v-u_j)| \le |A_j|/\delta_j$. Each factor's series is the exponential series, absolutely convergent with sum $\exp(|A_j|/\delta_j)$; the product of $2M$ absolutely convergent series is absolutely convergent with the product bound (24) (Mertens/Fubini for nonnegative terms), and the bound is uniform in v because δ_j does not depend on v. This is the explicit Leibniz/binomial regrouping: expanding the product of exponentials into a sum over $\mathbf{n}$ and collecting is legitimate precisely because the double (multi) family is absolutely summable. $\square$

Theorem 10.2 (Equivalence of the three price series)**.** *Let $\sigma_T^2 > 0$ and $\operatorname{Im} u_j < 0$ for all j (Lemma 8.2). Then the call price $C_M(K,T)$ of Theorem 8.3 admits three representations, all equal:*

(F1) ***Pole–residue / Schwinger–Gauss–erfc–Hermite form.***

$$C_M = S_0(1-e^k)^+ + \frac{S_0}{2\pi}\sum_{\mathbf{n}\in\mathbb{N}^{2M}}\Big(\prod_j \frac{A_j^{n_j}}{n_j!}\Big)\mathcal{I}_{\mathbf{n}}, \quad \mathcal{I}_{\mathbf{n}} := \int_{\mathcal{L}} \frac{e^{-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v} e^{-\mathrm{i}vk}}{v(v+\mathrm{i})\prod_j (v-u_j)^{n_j}}\mathrm{d}v, \tag{25}$$

each $\mathcal{I}_{\mathbf{n}}$ a finite $\mathbb{Z}$-combination of Gaussian-erfc kernel values $K_n(\zeta_\bullet, \sigma_T)$ of Lemma 9.2; the series converges absolutely. Grouping by a single pole gives the patent double sum $\sum_{j=1}^{2M}\sum_{n\ge 0}\frac{A_j^n}{n!}K_n(\zeta_j,\sigma_T)$-type form.

(F2) ***Single-index cumulant form.*** *With $\Psi(v) = \exp\sum_{m\ge 1} r_m v^m$ locally and Taylor coefficients q_N of Ψ satisfying the exponential-formula recurrence*

$$N\, q_N = \sum_{m=1}^{N} m\, r_m\, q_{N-m}, \qquad q_0 = 1, \tag{26}$$

the price is the single sum $C_M = \sum_{N\ge 0} q_N J_N$ with $J_N = \frac{S_0}{2\pi}\int_{\mathcal{L}} v^N e^{-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v} e^{-\mathrm{i}vk}/(v(v+\mathrm{i}))\mathrm{d}v$, convergent under the w-plane majorant $\|\Psi\|_R\,(G_0+G_1)/(1-R^{-1})$ with $R > 1$.

(F3) ***Edgeworth–Hermite form.*** *Let m_1 and κ_2 be the mean and variance of f_M and let $N(m_1, \kappa_2)$ be the variance-matched reference Gaussian. With standardized cumulants S_n of f_M and Gram–Charlier coefficients c_n from the Blinnikov–Moessner recurrence $n\,c_n = \sum_{k=1}^{n} k\,S_k\,c_{n-k}$,*

$$C_M = \mathrm{BS}(\sqrt{\kappa_2}, m_1) + S_0 \sum_{n\geq 3} c_n\,\Delta C_n(k), \qquad \Delta C_n(k) = \int_{\mathbb{R}} (e^{x+k} - e^k)^+\,\varphi_{m_1,\kappa_2}(x)\,\mathrm{He}_n(z)\,\mathrm{d}x, \tag{27}$$

$z = (x - m_1)/\sqrt{\kappa_2}$, convergent under the Cramér condition of Theorem 11.3.

All three equal C_M.

Proof. (F1). Insert (23) into (17) via (22). By Lemma 10.1 the multi-series converges uniformly on $\mathcal{L}$ and is dominated by e^{C_T}; the kernel $e^{-\frac{1}{2}\sigma_T^2 v^2}/(v(v+\mathrm{i}))$ is absolutely integrable on $\mathcal{L}$ (Lemma 8.4). Hence the product is dominated by an integrable function and term-by-term integration (dominated convergence) gives (25). Each $\mathcal{I}_{\mathbf{n}}$ is a Gaussian integral against a finite product of simple poles; partial-fractioning $1/(v(v+\mathrm{i})\prod_j (v-u_j)^{n_j})$ and applying Lemma 9.3 to each pole power expresses $\mathcal{I}_{\mathbf{n}}$ as a finite integer combination of kernel values $K_n(\zeta_\bullet, \sigma_T)$. Absolute convergence of the $\mathbf{n}$-sum follows from (24) times the uniform kernel bound. Grouping the absolutely summable family by the index of a single pole (all other $n_i = 0$) and collecting the analytic cofactor into the kernel argument yields the patent's by-pole double sum; absolute summability guarantees the grouped sum equals the ungrouped one.

(F2). On the disk $|v| < \min_j |u_j|$, $\varrho_M(v) = \sum_{m\geq 1} r_m v^m$ with $r_m = -\sum_j A_j u_j^{-m-1}$, and $\Psi = \exp \varrho_M = \sum_N q_N v^N$. The identity $\Psi' = \varrho_M' \Psi$ equates coefficients to give $N q_N = \sum_m m\,r_m\,q_{N-m}$, which is (26). This is the SAME exponential-formula recurrence proved for cumulant/moment passage in Section 4 (Newton's identity). Under the $w = \mathrm{i}v$ normalization of Proposition 8.5, single-series-pricing's radius $R > 1$ makes $\sum_N q_N w^{-N}$ converge on the line $\operatorname{Re} w = c \in (1, p^*)$ with majorant $\|\Psi\|_R (G_0 + G_1)/(1 - R^{-1})$; term-by-term integration against the kernel gives $C_M = \sum_N q_N J_N$. Because $\sum_N q_N v^N$ and the multi-series (23) are two expansions of the *same* analytic Ψ, and each is integrated against the same kernel with a dominating majorant, both integrals equal $\frac{S_0}{2\pi} \int_{\mathcal{L}} \varphi_M e^{-\mathrm{i}vk}/(v(v+\mathrm{i}))\mathrm{d}v = C_M$.

(F3). The inverse transform $f_M = \mathcal{F}^{-1}\varphi_M$ is a probability density with cumulant generating function $\log \varphi_M(-\mathrm{i}s) = \kappa_M(T, -\mathrm{i}s)$; its standardized cumulants S_n feed the Gram–Charlier/Edgeworth expansion $f_M(x) = \varphi_{m_1,\kappa_2}(x)(1 + \sum_{n\geq 3} c_n \mathrm{He}_n(z))$, $z = (x - m_1)/\sqrt{\kappa_2}$, with c_n generated by $nc_n = \sum_k kS_k c_{n-k}$ (Blinnikov–Moessner) and $S_1 = 0$, $S_2 = 1$ after variance matching. This recurrence is identical to (26) with $r_m \leftrightarrow S_m$ (Proposition 10.3 below), so the c_n are the same coefficients re-expressed in the Hermite basis. Pairing f_M with the call payoff (Parseval, Theorem 8.3) and integrating the Hermite series term-by-term—justified under the Cramér condition of Theorem 11.3—gives (27), where the $n = 0$ term is the Black–Scholes price $\mathrm{BS}(\sqrt{\kappa_2}, m_1)$ at the matched variance and $n = 1, 2$ vanish by the mean/variance matching. Since this is again the same Parseval integral $\frac{S_0}{2\pi} \int_{\mathcal{L}} \varphi_M e^{-\mathrm{i}vk}/(v(v+\mathrm{i}))\mathrm{d}v$, it equals C_M.

All three are term-by-term integrations of one integrand *kernel* $\times \Psi$ under convergent expansions of the single analytic function Ψ; each equals the Lewis integral C_M. Hence (F1), (F2), (F3) all equal C_M. ☐

Proposition 10.3 (Newton ≡ Blinnikov–Moessner)**.** *For a formal series $\exp(\sum_{m\geq 1} c_m v^m) = \sum_{N\geq 0} \Lambda_N v^N$ one has $N\Lambda_N = \sum_{m=1}^{N} m\,c_m\,\Lambda_{N-m}$; equivalently, the Blinnikov–Moessner recurrence $n\,c_n = \sum_{k=1}^{n} k\,S_k\,c_{n-k}$ recovers the Gram–Charlier coefficients from the standardized cumulants S_k. The two are the same identity.*

Proof. Differentiate $G(v) = \exp(\sum_m c_m v^m)$: $G' = (\sum_m m c_m v^{m-1})G$. Equate the coefficient of v^{N-1} on both sides: $N\Lambda_N = \sum_{m\geq 1} m c_m \Lambda_{N-m}$. Setting $c_m = S_m$ and reading $\Lambda_N = c_N$ (the Gram–Charlier weight generated by the same exponential) is the Blinnikov–Moessner form. Verified symbolically to high order. □

Provenance. Form (F1) is § F_1 of `patentApplication.tex` and `SchwingerHermiteFromCitations.tex`. Form (F2) is `single-series-pricing.md` (its majorant $\|\Psi\|_R(G_0 + G_1)/(1 - R^{-1})$). Form (F3) is what `RoughHestonEdgeworthCallPrice` compiles and is stated in `coefficient_algebra.tex`. The claim "the erfc–Schwinger–Gauss–Hermite closed form is the same as the Edgeworth expansion" is thereby *proved correct*: both are rearrangements of the one absolutely summable family (23), integrated against the same Lewis kernel. The Blinnikov–Moessner ≡ Newton identity (Proposition 10.3) closes the last gap between the two coefficient recurrences.

11 Infinite-interval Hermite validity: the exact Cramér inequality

The Hermite/Edgeworth form (F3) is an expansion on all of $\mathbb{R}$; its validity is governed by an exact inequality, not by any maturity restriction.

Proposition 11.1 (Hermite completeness on $\mathbb{R}$)**.** *The normalized Hermite functions $h_n(z) = (n!)^{-1/2}\mathrm{He}_n(z)$ are a complete orthonormal system of $L^2(\mathbb{R}, \varphi)$, φ the standard Gaussian weight: $\int_{\mathbb{R}} \mathrm{He}_m \mathrm{He}_n \, \varphi = n! \, \delta_{mn}$, and the closed linear span is all of $L^2(\mathbb{R}, \varphi)$. Consequently $g \in L^2(\mathbb{R}, \varphi)$ iff $g = \sum_n c_n \mathrm{He}_n$ with $\sum_n n! \, c_n^2 = \|g\|^2_{L^2(\varphi)} < \infty$, the series converging in $L^2(\varphi)$.*

Proof. Orthogonality is the classical Hermite relation (integration by parts on the Rodrigues formula). Completeness: if $g \in L^2(\varphi)$ is orthogonal to every He_n, then all moments of $g\varphi$ vanish; since the Gaussian weight has a finite-abscissa moment generating function, the moment problem is determinate and $g\varphi \equiv 0$, so $g \equiv 0$. Parseval gives $\sum_n n! \, c_n^2 = \|g\|^2$. □

Theorem 11.2 (Gaussian tail of the rational law)**.** *Assume* (C1)*: no pole $u_j(T)$ is purely imaginary, i.e. $\mathrm{Re}\, u_j \neq 0$ for all j. Then the moment generating function $M(s) = \varphi_M(-\mathrm{i}s, T) = \exp \kappa_M(T, -\mathrm{i}s)$ is finite for every real s and grows as $M(s) = \exp(\frac{1}{2}\sigma_T^2 s^2 - \mu_T s + O(1))$; equivalently f_M is sub-Gaussian with tail parameter σ_T^2:*

$$\limsup_{|x|\to\infty} \frac{-\log f_M(x)}{x^2/(2\sigma_T^2)} = 1. \tag{28}$$

Condition (C1) *is checkable at run time by testing $\mathrm{Re}\, u_j \neq 0$ for the computed poles.*

Proof. $\kappa_M(T, -\mathrm{i}s) = \frac{1}{2}\sigma_T^2 s^2 - \mu_T s + \varrho_M(-\mathrm{i}s)$. The proper part $\varrho_M(-\mathrm{i}s) = \sum_j A_j/(-\mathrm{i}s - u_j)$ has denominators $|-\mathrm{i}s - u_j|^2 = (\mathrm{Re}\, u_j)^2 + (s + \mathrm{Im}\, u_j)^2 \geq (\mathrm{Re}\, u_j)^2 > 0$ for all real s under (C1); hence $|\varrho_M(-\mathrm{i}s)| \leq \sum_j |A_j|/|\mathrm{Re}\, u_j| =: C'_T < \infty$ uniformly in s, and $\varrho_M(-\mathrm{i}s) \to 0$ as $|s| \to \infty$. Thus $M(s)$ is finite for all real s and equals $\exp(\frac{1}{2}\sigma_T^2 s^2 - \mu_T s + O(1))$. Sub-Gaussianity (28) follows from the Chernoff bound $P(X > x) \leq e^{-sx}M(s) = \exp(\frac{1}{2}\sigma_T^2 s^2 - (x + \mu_T)s + O(1))$ optimized at $s^* = (x + \mu_T)/\sigma_T^2$, giving $-\log P(X > x) \sim x^2/(2\sigma_T^2)$, and the matching lower bound from the saddle-point evaluation of $f_M = \mathcal{F}^{-1}\varphi_M$ (the saddle $v^* = -\mathrm{i}x/\sigma_T^2$ yields $-\log f_M(x) \sim x^2/(2\sigma_T^2)$). □

Theorem 11.3 (Exact Cramér inequality; infinite-interval validity for all T)**.** *Assume* (C1)*. Let $\kappa_2^{(M)}(T) = \mathrm{Var}(f_M) = \partial_s^2 \kappa_M(T, -\mathrm{i}s)\big|_{s=0}$ be the second cumulant and σ_T^2 the Gaussian-envelope parameter of Definition 8.1. Then, expanding around the variance-matched Gaussian $N(m_1, \kappa_2^{(M)})$:*

(a) *the Gram–Charlier/Edgeworth series* (F3) *converges in* $L^2(\varphi_{m_1,\kappa_2^{(M)}})$*, equivalently* $\sum_n n!\,c_n^2 < \infty$*, if and only if*

$$\boxed{\sigma_T^2 < 2\,\kappa_2^{(M)}(T)} \qquad \textit{(for every } T \textit{ in the model domain);} \tag{29}$$

(b) *under* (29) *the integrated price series* $S_0 \sum_{n\ge 3} c_n \Delta C_n(k)$ *converges* absolutely*, with*

$$\sum_{n\ge 3} |c_n|\,|\Delta C_n(k)| \le \Big(\sum_n n!\,c_n^2\Big)^{1/2} \Big(\sum_n \tfrac{1}{n!}\|\Pi_k\|_n^2\Big)^{1/2} = \|g\|_{L^2(\varphi)}\,\|\Pi_k\|_{L^2(\varphi)} < \infty, \tag{30}$$

where $\Pi_k(x) = (e^{x+k} - e^k)^+$ *is the payoff and* $g = f_M/\varphi_{m_1,\kappa_2^{(M)}} - 1$*.*

The inequality (29) *is exact and checkable at run time for each requested* T*; it is not an asymptotic or maturity-restricted statement.*

Proof. (a) By Proposition 11.1, (F3) converges in $L^2(\varphi_{\mathrm{ref}})$ iff $g = f_M/\varphi_{\mathrm{ref}} - 1 \in L^2(\varphi_{\mathrm{ref}})$, i.e. iff $\int_{\mathbb{R}} f_M^2/\varphi_{\mathrm{ref}} < \infty$ with $\varphi_{\mathrm{ref}} = \varphi_{m_1,\kappa_2^{(M)}}$. By Theorem 11.2, $f_M(x)^2 \asymp \exp(-x^2/\sigma_T^2)$ in the tail and $\varphi_{\mathrm{ref}}(x) \asymp \exp(-x^2/(2\kappa_2^{(M)}))$, so the integrand behaves as $\exp\big((-1/\sigma_T^2 + 1/(2\kappa_2^{(M)}))x^2\big)$; the integral is finite iff the exponent coefficient is negative, i.e. iff $1/\sigma_T^2 > 1/(2\kappa_2^{(M)})$, which is (29). This holds or fails by the value of the exact ratio $\sigma_T^2/\kappa_2^{(M)}(T)$ at the given T, with no smallness or maturity hypothesis.

(b) The payoff satisfies $\Pi_k \in L^2(\varphi_{\mathrm{ref}})$ unconditionally, since $\int_{\mathbb{R}} (e^{x+k})^2 \varphi_{\mathrm{ref}}(x)\mathrm{d}x = e^{2k}\,\mathbb{E}[e^{2X_{\mathrm{ref}}}] < \infty$ (a Gaussian exponential moment). Writing the Hermite coefficients of Π_k as $b_n = \frac{1}{n!}\langle \Pi_k, \mathrm{He}_n\rangle$, we have $\Delta C_n(k) = b_n \cdot n!$ up to the fixed normalization, and by Cauchy–Schwarz on the pairing $\langle g, \Pi_k\rangle_{L^2(\varphi)} = \sum_n n!\,c_n b_n$,

$$\sum_n |c_n||\Delta C_n(k)| \le \Big(\sum_n n!\,c_n^2\Big)^{1/2} \Big(\sum_n n!\,b_n^2\Big)^{1/2} = \|g\|_{L^2(\varphi)}\,\|\Pi_k\|_{L^2(\varphi)},$$

finite under (a). Hence absolute convergence of the price series. □

Corollary 11.4 (The closed forms are unconditional; only (F3) needs Cramér)**.** *Forms* (F1) *and* (F2) *converge to* C_M *under the sole hypotheses* $\sigma_T^2 > 0$ *and* $\operatorname{Im} u_j < 0$ *(Lemmas 8.2, 10.1); they do* not *require* (29)*. The exact price* C_M *is therefore always computable via* (F1)*. The Edgeworth form* (F3) *carries the additional exact hypothesis* (29)*, verified at run time; when it holds,* (F3) *equals* (F1)=(F2)= C_M*.*

Proof. Immediate from Lemma 10.1 (F1: absolute summability needs only $\delta_j > 0$) and Theorem 10.2 (F2: needs the envelope $\sigma_T^2 > 0$), versus Theorem 11.3 (F3: needs (29)). □

Remark (Convergence mode: L^2/pairing, not naive pointwise). The Edgeworth series (27) converges in $L^2(\varphi)$ and its *integrated* (price-paired) form converges absolutely; it is not asserted to converge pointwise as $\sum_n c_n \mathrm{He}_n(z)$ for fixed z, because $\mathrm{He}_n(z) \sim \sqrt{n!}\,e^{z^2/4}$ grows and the raw coefficient series can diverge pointwise while the L^2 and price-paired series converge. The pricing functional pairs g against the payoff $\Pi_k \in L^2(\varphi)$, so only the L^2 statement is needed and it is exactly (29).

Example 11.5 (Worked instance at $T = 1/12$; not the scope of the theorem)**.** For the parameters $\lambda = 0.1$, $\theta = 0.3156$, $\nu = 0.331$, $V_0 = 0.0392$, $\rho = -0.681$, $\mu = 0.62$, $\alpha \in (\frac{1}{2}, 1)$ (arXiv:2508.15080, eq. 1.3) at $T = 1/12$, the leading envelope $\sigma_T^2 = V_0 T \approx 3.27 \times 10^{-3}$ while the second cumulant

$\kappa_2^{(M)}(T) \approx 3.6 \times 10^{-3}$, so $\sigma_T^2 \approx 3.27 \times 10^{-3} < 2\kappa_2^{(M)} \approx 7.2 \times 10^{-3}$ with a comfortable margin (indeed $\sigma_T^2 < \kappa_2^{(M)}$). Thus (F3) converges here. This is one evaluation of the general Theorem 11.3, not its scope: the theorem holds for every T for which (29) is verified, checked at run time.

Provenance. The completeness/Cramér framework is `CONVERGENCE_PROOF.md` and the Hermite-validity discussion of `coefficient_algebra.tex`. The exact inequality (29) and its two-sided (iff) character, together with the separation of unconditional (F1)/(F2) from conditional (F3) (Corollary 11.4), are established here. Any framing of this as a "short-maturity" phenomenon is *incorrect*: the condition is the exact ratio $\sigma_T^2 < 2\kappa_2^{(M)}(T)$, valid and checkable for all T.

12 Reconciliation of the divergence document

We now adjudicate `EdgeworthDivergenceProof.tex` and the asymptotic-Edgeworth framing of the spectral-pricing document against Theorems 10.2 and 11.3. The resolution is a proved proposition, not prose: the two sets of results concern *different functions*, and we identify exactly which hypothesis separates them.

Definition 12.1 (True versus rational exponent)**.** Let $\Phi_\infty(v,T) = \log \mathbb{E}[e^{\mathrm{i}vX_T}]$ be the *true* rough-Heston characteristic exponent (the exact solution of the fractional Riccati equation), and $\kappa_M(T,v)$ the rational exponent of Definition 6.1. Write $s^*(T) = \sup\{s > 0 : \mathbb{E}[e^{sX_T}] < \infty\}$ for the critical moment of the true law.

Proposition 12.2 (The divergence thesis holds for the true exponent)**.** *For the true rough-Heston law, $s^*(T) < \infty$ (finite moment explosion). Consequently the true log-price density has a tail strictly heavier than every Gaussian, its standardized-cumulant Gram–Charlier series has coefficients c_n with $\sum_n n!\, c_n^2 = +\infty$, and the Edgeworth series about any fixed Gaussian* diverges *in L^2. Equivalently, the Cramér integral $\int f_\infty^2/\varphi_{\mathrm{ref}} = +\infty$ for every Gaussian reference.*

Proof. Rough Heston is an affine Volterra process whose moment generating function solves a fractional Riccati equation with a finite blow-up time in the s-variable; hence $s^*(T) < \infty$ (the same finite-critical-moment phenomenon as classical Heston, inherited through the affine structure). If $s^* < \infty$ then $\mathbb{E}[e^{sX_T}] = \infty$ for $s > s^*$, so the right tail of f_∞ satisfies $-\log f_\infty(x) = O(s^*x)$, i.e. at most exponential decay, strictly heavier than any $e^{-x^2/(2s^2)}$. Then for any Gaussian φ_{ref}, $\int f_\infty^2/\varphi_{\mathrm{ref}} \geq \int_{x>X_0} e^{-2s^*x} e^{+x^2/(2s^2)} \mathrm{d}x = +\infty$ because the Gaussian reciprocal grows super-exponentially. By Proposition 11.1 this is exactly $\sum_n n!\, c_n^2 = \infty$, so the Gram–Charlier series diverges in L^2. □

Proposition 12.3 (No contradiction: the pipeline prices the rational exponent)**.** *Theorem 11.3 concerns κ_M (rational, entire MGF under* (C1)*, Gaussian tail σ_T^2); Proposition 12.2 concerns Φ_∞ (finite critical moment, heavier-than-Gaussian tail). These are different functions, so their opposite convergence verdicts are not contradictory. Precisely, the hypothesis that fails for Φ_∞ is the Gaussian-tail premise of Theorem 11.2: Φ_∞ has a real MGF singularity at $s = s^*$ (its proper part is not a finite sum of poles off the imaginary axis but a transcendental function with a real-axis singularity), whereas κ_M satisfies* (C1) *and has an entire MGF. The rationalization step (diagonal Padé, Section 6) replaces the explosive tail by a Gaussian envelope, moving the MGF singularity off the real axis; this is precisely the regularization that makes* (29) *attainable.*

Proof. Immediate from Definition 12.1: the two theorems quantify over disjoint hypotheses. Theorem 11.2 requires (C1) ($\mathrm{Re}\, u_j \neq 0$, finitely many poles, entire MGF), which holds for κ_M and fails for Φ_∞ (whose MGF has the finite abscissa s^*). The pipeline evaluates the price of κ_M (the exact evaluation of the rational-exponent model to requested precision, Section 14), never of Φ_∞ directly; the difference $|\Phi_\infty - \kappa_M|$ is controlled by the Padé error (Theorem 7.2) and enters the ball radius, not the Cramér test. ☐

Proposition 12.4 (Status of the divergence document's internal arguments)**.** *In* `EdgeworthDivergenceProof.tex` *the § 5 Cramér-condition argument is* correct *and proves Proposition 12.2 for the true exponent; the § 3–4 steepest-descent/saddle argument is* incomplete *(a gap the document itself flags) and is not needed. The document's own concluding remark, that the Schwinger pole–residue series converges and is the production evaluation path, is* consistent *with Theorem 10.2 and Corollary 11.4. Likewise the spectral-pricing document's description of the Edgeworth series as* asymptotic *is* correct as stated for the true exponent *and does not conflict with the convergent* (F3) *for the rational exponent under* (29)*.*

Proof. The § 5 argument is the specialization of Theorem 11.3(a) to a law with finite critical moment: $\int f_\infty^2/\varphi = \infty$, hence divergence; this is valid. The § 3–4 saddle estimate attempts a pointwise asymptotic of c_n and does not close its remainder bound; since Proposition 12.2 is already established by § 5, the gap is immaterial to the thesis. The concluding remark and the spectral-pricing document "asymptotic" language both refer to the true exponent's series, whose divergence (as an L^2 series) coexists with its use as an *asymptotic* expansion; the rational exponent's (F3) is a distinct, genuinely convergent object by Theorem 11.3. No statement in either document contradicts the convergence theorems once the true/rational distinction of Definition 12.1 is applied. ☐

Provenance. `EdgeworthDivergenceProof.tex` (§§3–5) and the spectral-pricing document (asymptotic-Edgeworth framing). **Verdict:** the divergence *thesis is correct* for the true non-rational exponent (via the § 5 Cramér argument); the § 3–4 saddle argument is gappy but unnecessary; there is *no contradiction* with the convergence Theorem 11.3, which concerns the rational exponent κ_M under the run-time inequality (29). The divergence result is *irrelevant to the pipeline*, which prices κ_M, not Φ_∞.

13 Martingale identities, put–call parity, and the Black–Scholes limit

Theorem 13.1 (Martingale identities and put–call parity)**.** *For every M and T, $\kappa_M(T, 0) = \kappa_M(T, -\mathrm{i}) = 0$, hence $\varphi_M(0, T) = \varphi_M(-\mathrm{i}, T) = 1$. Consequently the call and put prices satisfy put–call parity*

$$C_M(K, T) - P_M(K, T) = S_0 - Ke^{-rT}. \tag{31}$$

Proof. The initial term of the recurrence is $a(0, v) = P(v)/\Gamma(\alpha + 1)$ with $P(v) = -\frac{1}{2}(v^2 + \mathrm{i}v) = -\frac{1}{2}v(v + \mathrm{i})$, whose zeros are exactly $v = 0$ and $v = -\mathrm{i}$. Fix $v_0 \in \{0, -\mathrm{i}\}$, so $a(0, v_0) = 0$. We show $a(k, v_0) = 0$ for all $k \geq 0$ by strong induction. The recurrence (7) reads

$$a(k, v_0) = Q(v_0)\, a(k-1, v_0)\, \tfrac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)} + R\, \tfrac{\Gamma((k-1)\alpha+1)}{\Gamma(k\alpha+1)} \sum_{j+m=k-2} a(j, v_0) a(m, v_0).$$

If $a(i, v_0) = 0$ for all $i < k$, both terms vanish, so $a(k, v_0) = 0$. Hence $a(k, v_0) \equiv 0$. Then $d(k, v_0) = 0$ for all k by (9), so the Padé numerator $P_M(z, v_0) \equiv 0$ and $\kappa_M(T, v_0) = V_0\Gamma(\alpha + 1)a(0, v_0)T + T\, P_M(T^\alpha, v_0)/Q_M(T^\alpha, v_0) = 0$. Therefore $\varphi_M(v_0, T) = e^0 = 1$. The identity $\varphi_M(0, T) = 1$ is the normalization of the law; $\varphi_M(-\mathrm{i}, T) = \mathbb{E}[e^{X_T}] = 1$ is the martingale property of the discounted asset. Put–call parity follows from the payoff identity $(e^x - \mathrm{k})^+ - (\mathrm{k} - e^x)^+ = e^x - \mathrm{k}$ under discounted expectation: $C_M - P_M = e^{-rT}(\mathbb{E}[S_T] - K) = S_0 - Ke^{-rT}$, using $e^{-rT}\mathbb{E}[S_T] = S_0\varphi_M(-\mathrm{i}, T) = S_0$. □

Provenance. The identities $\varphi(0) = \varphi(-\mathrm{i}) = 1$ are stated in `patentApplication.tex` (§on parity) and `notation.tex` (zeros of P). The induction $a(k, v_0) \equiv 0$ and its propagation to κ_M is supplied here; both source statements are *correct*.

Theorem 13.2 (Black–Scholes limit certificate)**.** *If the proper rational part vanishes, $\varrho_M \equiv 0$, then $\varphi_M(v, T) = \exp(-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v)$ is exactly Gaussian and the Lewis price (17) collapses to the exact Black–Scholes value*

$$C_M = S_0\,\mathcal{N}(d_1) - Ke^{-rT}\,\mathcal{N}(d_2), \qquad d_{1,2} = \frac{\log(S_0/K) + (r \pm \frac{1}{2}\sigma_{\mathrm{BS}}^2)T}{\sigma_{\mathrm{BS}}\sqrt{T}}, \tag{32}$$

with $\sigma_{\mathrm{BS}}^2 = \sigma_T^2/T$. This is a statement about the vanishing of the proper rational part *ϱ_M, not about the correlation parameter.*

Proof. If $\varrho_M \equiv 0$ then $\Psi \equiv 1$ and $\varphi_M = e^{-\frac{1}{2}\sigma_T^2 v^2 - \mathrm{i}\mu_T v}$ is the characteristic function of $N(\mu_T, \sigma_T^2)$. Substituting into (17) and evaluating the Gaussian integral by completing the square (equivalently, recognizing the Lewis integral of a Gaussian exponent) yields the classical Black–Scholes formula (32); the martingale identity $\varphi_M(-\mathrm{i}, T) = 1$ fixes the drift μ_T to the risk-neutral value so that the forward is $S_0 e^{rT}$. All (F1)–(F3) corrections vanish since $c_n = 0$ for $n \geq 3$ and the multi-index sum (23) reduces to its $\mathbf{n} = \mathbf{0}$ term. □

Provenance. `patentApplication.tex` states "$\rho \equiv 0$ collapses to Black–Scholes." **Verdict / disambiguation:** the correct hypothesis is the vanishing of the proper *rational part* $\varrho_M \equiv 0$ (the symbol ϱ/"$\rho(u)$" of the exponent decomposition), *not* the vanishing of the *correlation* $\rho = -0.681$. This resolves the ρ notation collision (Table 1): correlation and proper-rational-part share a glyph in the sources but are different objects; the Black–Scholes collapse is the rational-part statement.

14 The end-to-end arbitrary-bit certified enclosure

Theorem 14.1 (Arbitrary-bit certified price enclosure)**.** *Fix model data (Definition 1.1), a maturity T and strike K, and a requested bit count b. Assume* (C1) *and, for the Edgeworth route, the run-time inequality* (29)*. Then there exist a Padé order M, a series order N_H, and a working precision p, each computable from b and the data, such that the pipeline*

Müntz recurrence (7) $\to$ *d-sequence* (9) $\to$ *Chebyshev–Wheeler* (13) $\to$ $[M/M]$ *Padé* $\to$ *PFE* (16) $\to$ *price series* (

executed in outward-rounded ball arithmetic returns a ball $[C]$ with $C_\infty(K, T) \in [C]$ and radius $r([C]) \leq 2^{-b}$, where C_∞ is the exact call price. The radius decomposes as

$$r([C]) \leq \underbrace{\varepsilon_{\mathrm{P}}(M)}_{\textit{Páde}} + \underbrace{\varepsilon_{\mathrm{S}}(N_H)}_{\textit{series remainder}} + \underbrace{\varepsilon_{\mathrm{A}}(p)}_{\textit{arithmetic}}, \tag{33}$$

with $\varepsilon_{\mathrm{P}}(M) \leq L\,|\Delta_M|\,q_M/(1-q_M)$, $q_M \to e^{-2G} < 1$ (Theorem 7.2); $\varepsilon_{\mathrm{S}}(N_H) \leq \frac{S_0}{2\pi}e^{C_T}\sum_{|\mathbf{n}|>N_H}\prod_j(|A_j|/\delta_j)^{n_j}/n_j! \to 0$ super-exponentially (Lemma 10.1); and $\varepsilon_{\mathrm{A}}(p) \leq \mathcal{O}(\mathrm{ops})\,2^{-p}$ from outward rounding. The partial-fraction step is exact in rational arithmetic and contributes only to ε_{A}.

Proof. Enclosure. Every stage is executed in ball arithmetic with outward rounding, so each intermediate ball rigorously contains the exact intermediate value; by inclusion isotonicity the terminal ball $[C]$ contains the exact price of the rational-exponent model, and the Padé term ε_{P} additionally encloses C_∞ (true price) because $|C_\infty - C_M| \le L \sup_{\mathcal{L}} |\Phi_\infty - \kappa_M|$ with $L = \frac{S_0}{2\pi}\int_{\mathcal{L}} |v(v+\mathrm{i})|^{-1} e^{-\frac{1}{2}\sigma_T^2 u^2}\mathrm{d}u < \infty$ the Lewis-kernel Lipschitz constant, and $\sup_{\mathcal{L}} |\Phi_\infty - \kappa_M| \le |\Delta_M| q_M/(1-q_M)$ by Theorem 7.2 and the Stahl convergence of Theorem 6.5 (which guarantees $|\Delta_M| \to 0$ geometrically at rate e^{-2G}, so the target is attainable).

Budget. The three contributions are the only sources of error: the Padé order induces $\varepsilon_{\mathrm{P}}(M)$; the series order induces the tail $\varepsilon_{\mathrm{S}}(N_H)$ bounded by the master majorant (24) whose multi-index tail decays super-exponentially in N_H (factorial denominators); the finite-precision ball operations induce $\varepsilon_{\mathrm{A}}(p)$, linear in 2^{-p} with a constant counting the operations (all finite and data-determined). Their sum bounds $r([C])$ by subadditivity of ball radii under $+$.

Choice of parameters. Given b, pick M so $\varepsilon_{\mathrm{P}}(M) \le 2^{-b}/3$ (possible since $|\Delta_M| q_M/(1-q_M) \to 0$); pick N_H so $\varepsilon_{\mathrm{S}}(N_H) \le 2^{-b}/3$ (possible by super-exponential decay); pick p so $\varepsilon_{\mathrm{A}}(p) \le 2^{-b}/3$ (possible since it is $\mathcal{O}(2^{-p})$ with a fixed op-count once M, N_H are fixed). Then $r([C]) \le 2^{-b}$. □

Provenance. The three-part budget mirrors § I of `patentApplication.tex` ("ball radius = Páde error + statistical"); here the "statistical" contribution is identified with the series remainder ε_{S} plus the arithmetic term ε_{A}, and the Páde error is the rigorous a-posteriori bound of Theorem 7.2 (not the heuristic η_M^{stop}; Corollary 7.3). The claim "agrees with the solution for an arbitrary number of desired bits" is thereby *proved* as (33) with the constructive parameter choice.

A Conflict adjudication and notation crosswalk

The two propositions below are the formal adjudications forward-referenced from Sections 3 and 6; the tables collect every cross-document conflict with its verdict.

Proposition A.1 (Degree adjudication)**.** *In 0-indexed convention* $\deg_v a(k,v) = k+2$ *and* $\deg_v d(k,v) = k+3$*; in 1-indexed convention* $a_k = a(k-1,\cdot)$ *so* $\deg_v a_k = k+1$ *and* $\deg_v d_k = k+2$*. Hence:*

(a) the "$\deg a_k \le k+1$*" of* `patentApplication.tex/FrmpRoughHestonApp.tex` *and the "*$\deg a(k,v) = k+2$*" of* `RiemannHilbertView/FINDINGS.md/notation.tex` *are the* same statement *in different indexing (both* correct*, equality holding);*

(b) the claims $\deg d_k = k+4$ *(*`RiemannHilbertView`*) and* $\deg_v d(k,v) = 2k+4$ *(*`SchwingerHermiteFromCitations`*) are* both wrong*; the correct degree is* $k+3$ *(0-indexed), as in* `FINDINGS.md`*.*

Proof. From Theorem 3.3, $\deg_v a(k,v) = k+2$: base $a(0,v) = P(v)/\Gamma(\alpha+1)$ has $\deg_v = 2$; the recurrence's quadratic term $\sum_{j+m=k-2} a(j,\cdot)a(m,\cdot)$ attains $\deg_v = (j+2)+(m+2) = k+2$ at $j+m = k-2$, and the linear term $Q(v)a(k-1,\cdot)$ attains $\deg_v = 1+(k+1) = k+2$; both agree, giving exactly $k+2$. By Corollary 3.4, $\deg_v d(k,v) = \max(\deg_v a(k,v), \deg_v a(k+1,v)) = \max(k+2, k+3) = k+3$. The reindex $a_k = a(k-1,\cdot)$ subtracts one from each. Verified symbolically to $k = 8$. □

Proposition A.2 (Müntz radius adjudication)**.** *The statements "the Müntz series is divergent" (*`FrmpRoughHestonApp.tex`*,* `CONVERGENCE_PROOF.md`*,* `HANDOFF.md`*) and "the Müntz series has a positive radius of convergence" (*`PadeMuntzSpectralTau`*,* `ThePadeMuntz`*) are* both correct*: they concern different variables. The series* $\sum_k d(k,v) z^k$ *in the Padé variable* $z = T^\alpha$ *has positive radius* $\rho_0(v)$ *(Lemma 4.3); the formal Puiseux/asymptotic series in* t *at fixed* t *diverges (it is an* *asymptotic, not convergent, expansion). The Padé step resums the z-series (Proposition 6.2); there is no conflict.*

Proof. Lemma 4.3 gives a majorant geometric series with ratio bounded by $c_1\rho_0 + c_2\rho_0^2 < 1$, hence $\rho_0(v) > 0$ and absolute convergence of the z-series on $|z| < \rho_0$. The t-series is the reordering by powers of t of the Müntz monomials $t^{k\alpha}$; its coefficients grow factorially (the fractional Riccati solution is not entire in t), so it is a genuine asymptotic series of radius 0 in t. Different variables, different convergence type; both statements hold as stated. $\square$

Conflict adjudication table (document × claim × verdict)

Topic	Conflicting claims	Verdict / correction
$\deg a_k$	patent/FRMP: $\deg a_k \le k+1$ (1-idx) *vs.* RiemannHilbert/FINDINGS/notation: $\deg a(k,v) = k+2$ (0-idx)	**Both correct**, pure indexing offset $a(k)_0 = a_{k+1}$; equality holds (Prop. A.1). Verified to $k=8$.
$\deg d_k$	RiemannHilbert: $\deg d_k = k+4$ *vs.* Schwinger: $\deg_v d(k,v) = 2k+4$	**Both wrong**; correct $\deg d(k,v) = k+3$ (0-idx), FINDINGS.md is right (Prop. A.1, Cor. 3.4).
Müntz radius	PadeMuntz: positive radius *vs.* FRMP/CONVERGENCE: divergent series	**Both correct**: $z = T^{\alpha}$-series converges ($\rho_0 > 0$); t-series is asymptotic. Padé resums the z-series (Prop. A.2).
Schwinger correction	patent §F$_1$: $-Q_n/\sigma_T$ *vs.* notation: $F^{(n)} = P_nF - \frac{\sqrt{2}}{\sqrt{\pi}\sigma}Q_n$	**Patent typo**: correct exponent is $-Q_n/\sigma^2$ (Lemma 9.2); the P/Q recurrence itself is correct. Verified symbolically ($-Q_n/\sigma$ fails at $n=1$).
Contour convention	PricingTheorem/CONTOUR: v-plane $c \in (-1,0)$ *vs.* single-series: w-plane $c \in (1,p^*)$	**Same integral** under $v = -\mathrm{i}w$; boundary terms differ by which kernel poles the line separates (Prop. 8.5).
FRHA d_k	CROSS_DOCUMENT_AUDIT: FRHA uses $a(k-1)$ (wrong) *vs.* current FRHA: uses $a(k+1,v)$	**Audit finding stale**: current `FrmpRoughHestonApp.tex` uses $a(k+1,v)$, matching notation/patent. FRHA now correct.
Edgeworth diverge/converge	EdgeworthDivergence: price series diverges *vs.* patent/CONVERGENCE: Schwinger series converges	**No contradiction**: divergence is for the *true* exponent (moment explosion); convergence is for the *rational* κ_M (Prop. 12.3).
Divergence-doc soundness	EdgeworthDivergence §3–4 saddle bound *vs.* its own §5 Cramér proof	§3–4 **self-admitted gap**; §5 Cramér argument **correct and complete**. Thesis holds via §5 (Prop. 12.4).
σ_T^2 meaning	PricingTheorem: $\sigma_T^2 = V_0T + \frac{1}{2}\lambda\theta T^2 + \dots$ *vs.* Edgeworth: $\sigma_T^2 = \kappa_2(T)$	σ_T^2 is the Gaussian **envelope** (polynomial part of κ_M); κ_2 is the **variance**; they differ. Cramér: $\sigma_T^2 < 2\kappa_2$ (Def. 8.1, Thm. 11.3).
Primary route	CROSS_DOCUMENT_AUDIT: Padé-in-z primary, residue cross-check *vs.* patent: Schwinger residue preferred	**Reconciled**: Padé-in-z resums the cgf; residue form is exact evaluation of the fixed-M Lewis integral. Same C_M (Thm. 10.2).

Topic	Conflicting claims	Verdict / correction
ρ collision	correlation $\rho = -0.681$ *vs.* proper rational part $\rho(u) = \varrho_M$	Distinct objects sharing a glyph. Black–Scholes collapse is $\varrho_M \equiv 0$ (rational part), **not** correlation $= 0$ (Thm. 13.2).
c_j collision	Edgeworth weights c_j *vs.* residues c_j *vs.* contour abscissa c	Three distinct roles; disambiguated in Table 1. Edgeworth c_n via Blinnikov–Moessner $\equiv$ Newton (Prop. 10.3).
CROSS_DOCUMENT_AUDIT	its own conclusions	**Mostly correct**; the FRHA off-by-one finding is **stale** (row 6). Its route-primacy framing is reconciled (row 10). Verified independently.

Notation crosswalk

Symbol	Meaning in this document	Source aliases
α	fractional order, $\alpha = H + \frac{1}{2} \in (\frac{1}{2}, 1)$	H Hurst index
v	Fourier/pricing variable (v-plane)	u (patent), $w = \mathrm{i}v$ (single-series)
σ_T^2	Gaussian envelope (polynomial part of $-\kappa_M$)	"σ_T^2" (PricingTheorem)
$\kappa_2^{(M)}(T)$	second cumulant / variance of f_M	"κ_2", "σ_T^2" (Edgeworth doc, conflated)
$\varrho_M(v)$	proper rational part $\sum_j A_j/(v - u_j)$	"$\rho(u)$" (patent), "proper part"
ρ	correlation $= -0.681$	(same glyph as ϱ_M in sources)
A_j, u_j	PFE residues and poles, $\operatorname{Im} u_j < 0$	"$A_j^n/n!$", "ζ_j"
c_n (Edgeworth)	Gram–Charlier weight, Blinnikov–Moessner	standardized-cumulant weight
c (contour)	abscissa of the pricing line	β, η_0 (strip offset)
K_n, P_n, Q_n	Schwinger kernel and its polynomials	$F^{(n)}, P_n, Q_n$ (notation.tex)
Δ_M	Padé/price increment $\kappa_M - \kappa_{M-1}$	$\lvert\Delta_M\rvert$ (Algorithm 4)
$z = T^\alpha$	Padé variable	"z", "t^α"
η_M^{stop}	heuristic stopping estimate	a-posteriori estimate (Alg. 4)

Remark (Summary verdict on the two prior audits). `CROSS_DOCUMENT_AUDIT.md` is *substantially correct* and its route-reconciliation is sound; its single defect is the stale FRHA off-by-one (now corrected in source). `EdgeworthDivergenceProof.tex` reaches a *correct thesis* for the true exponent through its §5 Cramér argument, with a *non-fatal gap* in its §3–4 saddle estimate; it poses *no* obstruction to the arbitrary-bit certified pricing of the rational-exponent model (Theorem 14.1).